

CIS 21031 – PRACTICAL FOR PLATFORM TECHNOLOGIES
DEPARTMENT OF ICT
FACULTY OF TECHNOLOGY
SOUTH EASTERN UNIVERSITY OF SRILANKA

Labsheet: 3

Date: 24th of November 2023

Title: Programming Shell Scripts-I

Aim: Getting practice to write and execute the shell scripts.

Task

- Create a shell script
- Use a text editor to put required Linux commands and logic in the file
- Make the script executable.

Exercise 01:

1. Open Terminal and navigate to Desktop directory using terminal commands.
2. Create a sub directory with the name **Day3** inside the desktop directory.
3. Create a shell script file called 'Exercise1.sh' inside the created sub that directory using Terminal command.
4. Open the file 'Exercise1.sh' using the sublime text editor and write your first Bourne Again Shell (BASH) shell script which display says, "**Hello, World!**".
5. Save and close the file.
6. Run the script using the following terminal command and get the output.
 - (**./Exercise1.sh**)

Exercise 02: Variables in Shell

1. Create a Bourne Again Shell (BASH) script 'Exercise2_1.sh' to display the following information based on System Variables in shell.
 - a) Print "Welcome [Logging Name] " (*Eg: Welcome root*)
 - b) Print "You are working on [your shell name] shell"
 - c) Next print "You are currently working on [Current Working Directory] directory".
 - d) Then print "You are using [Available OS] operating system.

e) Finally print “Bye!”

2. Create a Bourne Again Shell (BASH) script ‘Exercise2_2.sh’ to display the following information based on **User Defined Variables** in shell.
 - a) Create a variable called “Course”
 - b) Give the value “Platform Technology” to the variable you created.
 - c) Display the value of variable course with echo command.
 - d) Repeat the question in Exercise 02 part 1 by defining separate variable for each system variable to store their values and display the information given there.

Exercise 03: Array Variables

1. Create a Bourne Again Shell (BASH) script “Exercise03.sh” to display the following information based on Array variables in shell.
 - a) Create an array variable called “Days” and store four days of the week in the array. (Monday, Tuesday, Wednesday, Thursday)
 - b) Print the first element of the array.
 - c) Print the second element of the array.
 - d) Print all the elements in the array.
 - e) Add another element called “Friday” as the last element and print that element.
 - f) Change the value of the second element to “Sunday”.

Practice Question:

1. Execute the following terminal commands and discuss the output
 - a. useradd
 - b. passwd
 - c. userdel
 - d. groupadd
 - e. groupdel
 - f. usermod

2. Create a directory with your Registration Number inside home directory.
 - a. Create a shell script 'LabRecord3_2.sh' and do the following.
 - i. Print "CIS21031 LabSheet03 – Introduction to Shell Scripting"
 - ii. Modify the shell script to include a variable. The variable will hold the contents of message in **a - i.**
 - iii. Store the values 6, 6.75 and "Technology" respectively into the appropriate variables and display those values.
 - iv. Store the output of the command "hostname" in a variable. Display the script is running on _." where "_" is the output of the "hostname" command
3. Create a shell script 'LabRecord3_3.sh' and do the following.
 - i. Create an array variable called 'Fruits' and add four values to it (Apple, Orange, Grapes, Banana).
 - ii. Print the first element of the array.
 - iii. Print all the elements in the array
 - iv. Add another element called "Watermelon" as the last element and print that element.
 - v. Print the elements starting from index 2.
 - vi. Change the value of the second element to "Pineapple".
 - vii. Get the number of elements in the array.
 - viii. Delete last before element in the array.

Submit your answers on or before 1st of December 2023.